

NAME - Montu Jangid

Assignment – Introduction to Cybersecurity

Module 1 – Introduction to Cyber Security

1. What is the Meaning of Cyber Security?

Answer -

Cyber security is the process of protecting computers, networks, systems, and data from cyber attacks, unauthorized access, viruses, malware, and data theft.

It helps in keeping digital information safe and secure. Cyber security is used in banking, business, government, healthcare, and personal systems to protect sensitive information.

2. What are the Main Objectives of Cyber Security?

Answer -

The main objectives of cyber security are:

1. Confidentiality

Ensures that sensitive information is accessed only by authorized users.

2. Integrity

Ensures that data remains accurate and is not modified without permission.

3. Availability

Ensures that systems, applications, and data are available whenever required.

4. Authentication

Verifies the identity of users before providing access to systems.

5. Non-Repudiation

Prevents users from denying actions or transactions performed by them.

3. What is Offensive and Defensive in Cyber Security?

Answer -

Offensive Cyber Security

Offensive cyber security is the practice of testing and attacking systems legally to identify vulnerabilities and security weaknesses before real attackers exploit them.

Examples:

- Ethical hacking
 - Penetration testing
 - Vulnerability assessment
-

Defensive Cyber Security

Defensive cyber security focuses on protecting systems, networks, and data from cyber attacks. It involves monitoring, detection, and prevention of threats.

Examples:

- Firewalls

- Antivirus software
 - Intrusion detection systems
-

4. What is Cyberspace and Law?

Answer -

Cyberspace

Cyberspace is the virtual environment created by computer networks, the internet, software, and digital communication systems where online activities take place.

Examples include websites, email systems, social media, cloud computing, and online banking.

Cyber Law

Cyber law is a set of laws and regulations related to the use of computers, networks, digital communication, and the internet. It helps prevent cyber crimes and protects users' digital rights and information.

Cyber laws deal with:

- Hacking
 - Identity theft
 - Online fraud
 - Data theft
 - Cyber bullying
-

5. What is Cyber Welfare?

Answer -

Cyber welfare refers to the safe, secure, and responsible use of digital technologies and the internet for the benefit of individuals and society.

It focuses on spreading awareness about cyber security, protecting users from cyber crimes, and encouraging safe online behavior.

Objectives of Cyber Welfare:

- Safe internet usage
 - Protection of personal data
 - Digital awareness
 - Prevention of cyber crimes
-

6. Explain the Types of Hacker

Answer -

1. White Hat Hacker

White hat hackers are ethical hackers who legally test and improve system security.

2. Black Hat Hacker

Black hat hackers illegally access systems to steal data, spread malware, or damage systems.

3. Grey Hat Hacker

Grey hat hackers access systems without permission but usually do not have harmful intentions.

4. Script Kiddies

Script kiddies are inexperienced hackers who use pre-made hacking tools without deep technical knowledge.

5. Hacktivists

Hacktivists perform cyber attacks for political or social causes.

6. State-Sponsored Hackers

These hackers work for governments and perform cyber espionage or cyber warfare activities.

7. What is the Full Form of SOC in Cyber Security?

Answer -

SOC stands for **Security Operations Center**.

It is a centralized unit responsible for monitoring, detecting, analyzing, and responding to cyber security threats and incidents in an organization.

Functions of SOC:

- Continuous monitoring
 - Threat detection
 - Incident response
 - Security analysis
 - Log management
-

8. What are the Challenges of Cyber Security?

Answer -

Cyber security faces many challenges because cyber threats are increasing continuously.

Major Challenges of Cyber Security:

1. Increasing Cyber Attacks

Cyber attacks such as phishing, malware, and ransomware are growing rapidly.

2. Data Privacy and Protection

Protecting sensitive and personal information from unauthorized access is difficult.

3. Lack of Skilled Professionals

There is a shortage of trained cyber security experts.

4. Advanced Malware and Ransomware

Modern cyber threats are becoming more complex and difficult to detect.

5. Cloud Security Risks

Securing data stored in cloud environments is challenging.

6. Insider Threats

Employees or internal users may accidentally or intentionally compromise security.

7. Rapid Technological Changes

New technologies create new vulnerabilities and security risks.